

A novel 2.5bit SHA-less MDAC design for 10bit 100Ms pipeline ADC

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Abstract: A sub circuit block - 2.5bit MDAC used in 10 bit 100MHz pipeline ADC fabricated in TSMC 0.35um process is presented in this paper. The pipeline ADC removes a front Sample and Hold block (SH) to save power dissipation and area. However, the aperture error between the multiplying digital-to-analog converter (MDAC) and the comparators in the sub flash may increases in large amount. It obviously decreases the precision of sample phase. In this paper, there are two ways to decrease the aperture error. Furthermore, in order to improve the accuracy of sample signal, a simplified boost-strapped switch is adopted in first 2.5bit SHA-less MDAC design, which only uses one capacitor to restore the adding voltage. The result shows that the MDAC consume 30.73 mW under 3.3v power supply. The settling time is 3ns for input signal at Nyquist rate of 50mHz.

Key Words: Pipeline ADC; MDAC; SHA-less; boost-strapped switch

I. INTRODUCTION

Data conversion IC plays an important role in current wireless communications. Analog to digital conversion quantifies signal from the outer source. Many researches are engaged in obtaining high sampling rate and high resolution. In many applications, such as transceivers and video application, the pipeline ADC is suitable to meet strict requirement of system design. One trend of ADC revolution is to reduce the signal routine and power dissipation. To achieve this goal, removing front Sample and Hold block is a reproach for consideration, since the SH is not a necessary part to design pipeline ADC, instead, a big burden of power consumption. However, once SH is removed, there are other sampling errors encountered in the sampling network, such as mismatch and aperture errors which will deteriorate the performance of ADC. In this paper we present a reasonable system design to achieve the specification of the system on chip.

Firstly, a novel 2.5bit MDAC without SH for first stage dedicate in 3.3V 10bit 100MS/s CMOS pipeline ADC is demonstrated. Traditional SHA-less architecture MDAC require a few pre amplifiers for amplifying the difference between the reference voltage and input signal during the judge time. Actually if well designed, the pre amplifiers can be eliminated to save time. However it does not affect the correction status while the Cr is designed smaller to measure the difference faster. So latches should be designed faster than traditional structures. Carefully checking on timing

skew is required to guarantee the performance of the whole circuit.

The paper is organized as follows: Section two proposes the system definition of 10bit SHA-less ADC. Section three describes the first MDAC design. Section four reports the simulation results. And the last section summarizes the design flow.

II. SYSTEM DEFINITION OF SHA-LESS ADC

Speed, resolution and power consumption are three factors often being compromised in designing of pipeline ADC system. The faster the speed, the more errors in accuracy of sample signals, which will affect the resolution of the ADC. The more resolution, the more quantized stages are required, which will generate more power occupation. The bits get from each stages are also needed for compromise consideration, since the noise and gain errors in 4 or 5 bit MDAC stages may increase. It also need higher gain bandwidth OP to decrease the holding error during holding time. Sub flash ADC will need more numbers of comparators, which is not expected. So we choose first stage as 2.5 bit, which relaxes the requirement in designing residue OP and increase the accuracy.

A. Pipeline ADC with no Sample and Hold

Fig.1 shows the structure model of front SHA less ADC, which consists of several circuit blocks such as: quantizes stages that including first 2.5 bit MDAC with sub DAC, four stages 1.5bit MDACs and last 3bit flash. The clock receiver generates the sample and amplifies clock separately. Voltage bias of irrelative with temperature and voltage supply fluctuation is designed to supply power. Digital calibration makes quantized digital data correctly align to the output. The digital data should be largely current droved.

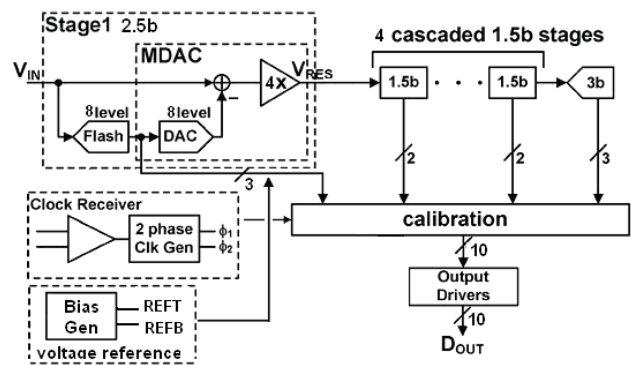


Figure 1. Structure of SHA less ADC with 10bit resolution

B. First 2.5bit stage MDAC

In Fig2, we use a multi-bit front-end to offer significant power savings for high-SNR noise-limited pipeline ADCs at the cost of increasing circuit complexity. The first stage of MDAC measures the difference between the input signal and reference voltage, and transfer the charge from C_s to the C_f , the output voltage is expressed in the following transfer function:

$$V_{out} = (1 - \frac{1}{A\beta}) \times \left[\left(\frac{C_s + C_f}{C_f} \right) \times V_{in} - D \times \left(\frac{1}{1 + \frac{1}{A\beta}} \right) \times V_{ref} \times \left(\frac{C_s}{C_f} \right) \right]$$

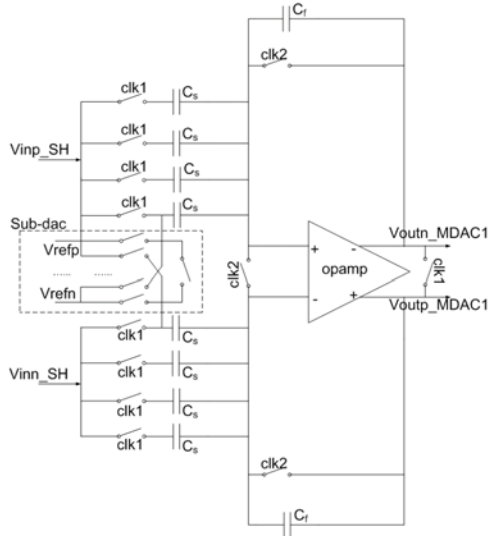


Figure 2. Traditional MDAC topology

The timing is shown in Fig 3. We separate the sample clock as ϕ_1 and ϕ_{11} , in which ϕ_{11} samples the input signal and accomplish encode in sub DAC during T latch, then go to amplify the residue, so at the sample phase ϕ_{22} is designed superior than ϕ_2 to obtain the small interval of T latch. Moreover, the boost-strapped switches are adopted to achieve higher linearity during sample phase. Plus, ϕ_{11P} is applied to execute under capacitor sample.

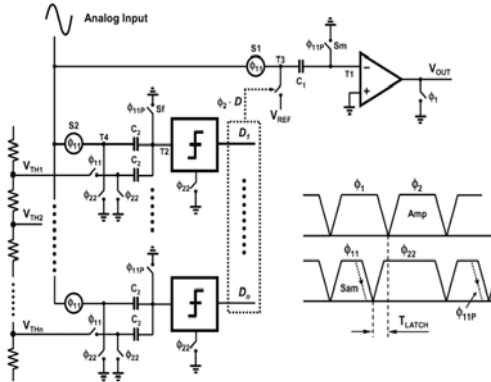


Figure 3. The first MDAC [5]

Typical transfer function of MDAC is follows as figure 4. The slope of the curve is 4. The correction range is $V_{ref} \times 2 \times 1/8$. Due to different digital calibration algorithm, the judge zone is also different at $(-1, -\frac{7}{8})$ and $(\frac{7}{8}, 1)$. It adds more 2 judgment levels than traditional transfer function.

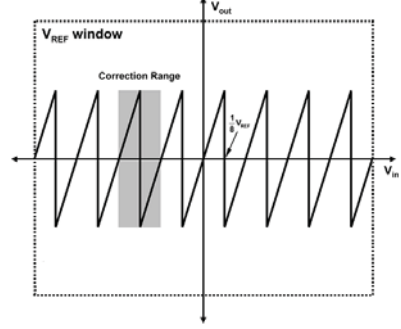


Figure 4 the transfer function of 2.5 bit MDAC

C. Booststrapped switch

The boost strapped switch, as shown in Fig.5, allows the output signal follow the input signal exactly. A simple design of boost strapped switch only uses one capacitor to realize the function, which saves areas on the chip than traditional type. The capacitor will occupy more areas on the chip, since the first stage sub ADC uses 8 compare levels. It will seriously consume more area than traditional CMOS switches.

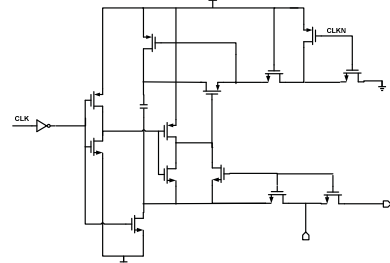


Figure 5. The boost strapped switch

D. Latch

Cross type of latch shown in figure 5 is commonly used, designed by much more size of each MOS in order to lock the voltage faster. In this case, we need the response time less than 0.3ns, and the latch will detect differential signals less than 5uV.

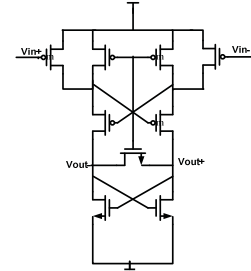


Figure 6. The fast locked latch

E. Residue OP

According to 10bit 100Ms pipeline ADC specifications, for the 2.5 bit MDAC at first stage, the gain of the residue OP increases by a factor of four and the feedback factor decreases by a factor of four. For analysis, close loop bandwidth (BW) of a stage amplifier is given by

$$BW = \beta \frac{g_m}{C_L}$$

(β is feedback factor, C_L is load capacitors, g_m is input CMOS Trans conductance).

In order to fulfill the resolution, the holding error is shown in the following formula:

$$\varepsilon = V_{swing} \times e^{-BW \times T_S} = V_{swing} \times e^{-T_S / \tau} \leq \frac{1}{2} \frac{V_{swing}}{2^{N-K+1}}$$

($K= [2.5]= 3$, Voltage Swings=2v, $N-K+1=N_{residue}$) With calculation, the error should be less than 0.976mV.

$$\tau = \frac{1}{\omega_u} < \frac{T_S}{(N_{res}+2)\ln 2}$$

We can deduce the unity bandwidth as:

$$f_u > \frac{(N_{res} + 2) \ln 2}{2\pi \times \beta \times T_S}$$

So settling time T_S should less than 5ns, unit BW f_u should larger than 967.6MHz.

Due to the requirement accuracy,

$$\frac{1}{A_0 \times \beta} < \frac{1}{4} \text{LSB}$$

DC Gain A_0 should larger than:

$$\frac{4}{\beta} \times 2^{N_{RESIDUE}}$$

As $\beta = \frac{1}{5}$, $N_{residue}$ is 8, the Gain should be larger

than 74.2dB. The actual open loop gain should larger than 80 dB.

In the 3.3v process, there is enough head room for cascode design, so the 2 stage cascode OP shown in fig.7 with miller compensation can implement the high gain and fast settling time. R_z and C_c can eliminate the zero at left half plane. CMFB circuit block is to stabilize the tail current.

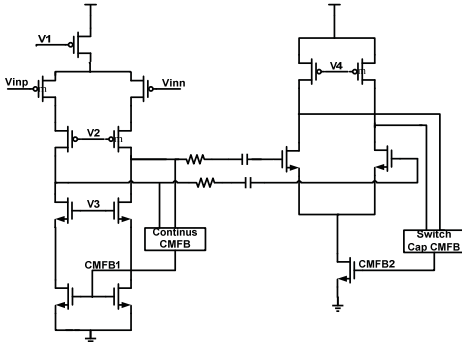


Figure 7.The residue 2 stage Operation Amplifier

III. SIMULATION RESULTS

A. Two stage OP simulation results

With the compensation $C_c=2.5\text{pF}$, $R_z=100\text{ohm}$, OP test results is shown as table 1. Its consumption is 26.4mW at TT corner.

Corners	Gain (dB)	GBW (GHz)	Phase Margin	Slew Rate(V/us)
TT	83.2	1.12	50.28	1200
SS_LVHT	71.4	1.05	48.46	860
FF_HVLT	78.6	0.97	43.04	940

Table 1.The residue OP parameters

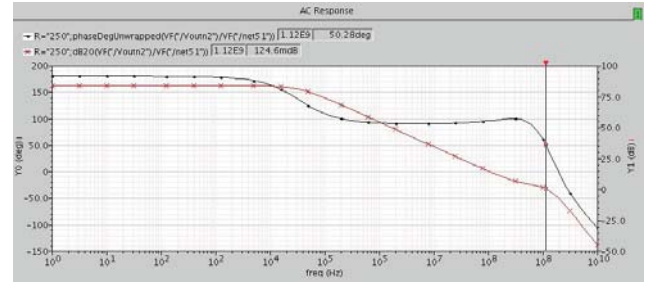


Figure 8.The residue OP parameters

B. Bootstrapped swiches test results

The simulation results of a sample and hold circuit show that THD (dBc) the simplified bootstrapped switch is 76.8 dBc.

C. First 2.5bit MDAC simulation results

The designed MDAC is simulated to operate with a maximum 2v input signal swing (dual) with a 3.3v power supply. The maximum residue is

$$\frac{\text{SignalSwing}}{\text{Compare levels}} \times \text{Gain} \times 2 = \frac{2V}{16} \times 4 = 0.5V$$

While the positive input is slope voltage from 1.3v to 2.3V, the negative input is from 2.3v to 1.3v; actual transfer function is simulated as in Figure. 8:

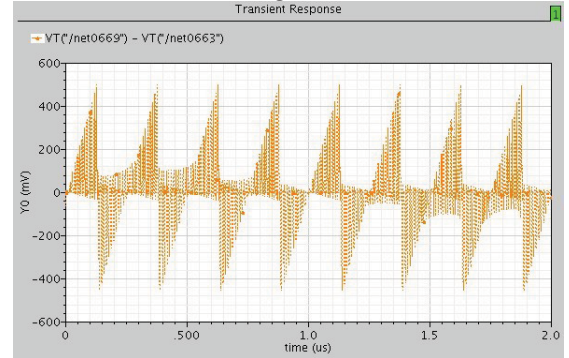


Figure 8.The pro-simulation transfer function

Fig 8 demonstrates that the sub-ADC have correctly judge the V_{ref} . As the capacitor is chosen smaller just 200fF, the latch will soon detect the difference voltage, the sub DAC also does the right encode.

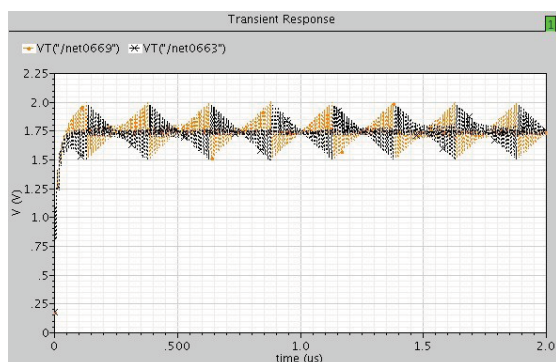


Figure 9. The differential of the output signal

We can see that the common mode output of the OP from figure 9 is 1.73V. And the rising time of common mode output is less than 0.2us, which means the stability of CMFB is well designed.

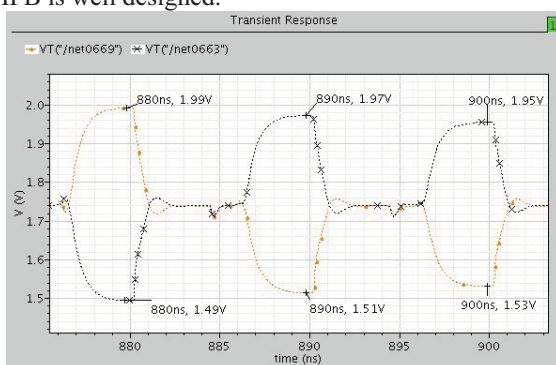


Figure 10. The detail of the output signal, while input is slope

From the simulation results, the settling time of the OP is 3.2ns less than 5ns; the gain error is 0.75mV less than 0.98mV.

IV. CONCLUSIONS

This work represents a new design of 2.5bit MDAC which is suitable for pipeline ADC of 10bit of resolution and 100M sample/s conversion rate in 0.35um CMOS process. The power consumption of this MDAC is 30.73 mw and its settling time is 3ns. A two stage amplifier with miller compensation is implemented to satisfy the strict design requirement. The pre-amplifier not only amplify the

signal but also amplify the inject charge from switch action. So the pre-amplifier is removed in the new proposal to reduce the affect. The latches keep fast response and sensitivities which can detect the input differences less than 5uV. To improve the accuracy of the sample signal, boost-strapped switch is adopted in this SHA-less MDAC design, which adds extra power dissipation. The pro simulation results of this work reveals that SHA-less architecture saves power consumption and realizes the traditional SH function. The only defect for this design is the switches occupy more area in silicon chip.

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